

DEPARTMENT OF MATHEMATICS & STATISTICS

MTH 211: THEORY OF STATISTICS

Session: 2025-26 Semester: (II)

End Semester Exam

Maximum Marks: 100

Time: 180 minutes

Important instructions (marks will be deducted, if not followed):

- (1) Start answering a new question in a new page.**
- (2) Answer all parts of a question together in one place.**
Leave adequate space to attempt any part later.
- (3) If there are multiple solutions of a problem in different places, only the first solution will be checked. So, strike out all except the one you wish to be checked.**

1. Let $X_1, \dots, X_n$ be a random sample from a Kumaraswamy distribution with pdf

$$f_X(x) = \theta x^{\theta-1} \beta (1-x^\theta)^{\beta-1}, \quad 0 < x < 1; \quad \theta \text{ (known)} > 0, \quad \beta > 0.$$

(a) Find a complete sufficient statistic for β .

(b) Find the MLE of β .

(5+5=10 marks)

2. Let $X_1, \dots, X_n$ be a random sample from a population with pdf

$$f_X(x) = \frac{\theta x^{\theta-1}}{3^\theta}, \quad 0 < x < 3; \quad \theta > 0.$$

(a) Find the Method of Moments Estimator (MoME) $\hat{\theta}_{Mo}$ of θ .

(b) Prove or disprove that $\hat{\theta}_{Mo}$ is consistent.

(c) Prove or disprove that $\hat{\theta}_{Mo}$ has an asymptotic normal distribution.

(2 + 3 + 5 = 10 marks)

3. Let $X_1, \dots, X_n$ be a random sample from a population with pdf

$$f_X(x) = \frac{\alpha}{\theta^\alpha} x^{\alpha-1}, \quad 0 \leq x \leq \theta; \quad \theta, \alpha > 0.$$

(a) Find the MLE of θ and α .

(b) If α is known, find a, b such that $(aX_{n:n}, bX_{n:n})$ is a 90% confidence interval for θ .

(6 + 6 = 12 marks)

4. Let $X_1, \dots, X_n$ be a random sample from a population with pdf

$$f_X(x) = \frac{2x}{\theta} \exp\left(-\frac{x^2}{\theta}\right), \quad x > 0; \quad \theta > 0.$$

(a) Find the CRLB for the variance of an unbiased estimator of θ .

(b) Find the UMVUE of θ .

(6 + 4 = 10 marks)

5. Let $X_1, \dots, X_n$ be a random sample from a $U(0, \theta)$, $\theta > 0$ distribution.
- (a) The prior of θ is $\beta\alpha^\beta\theta^{-(\beta+1)}$, $\alpha < \theta < \infty$; α, β known. Obtain the posterior mean.
- (b) The prior of θ is $\frac{1}{\theta\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln\theta-\mu)^2}{2\sigma^2}\right)$, $\theta > 0$; μ, σ known. Obtain the posterior mode.
- (6 + 6 = 12 marks)**

6. Let $X_1, \dots, X_n$ be a random sample from $Ber(p)$, with $P(X_i = 1) = p(\in (0, 1)), \forall i$.
- (a) Under the squared error loss, obtain the Bayes estimator of $p(1-p)$ when the prior is $Beta(\alpha, \beta)$, with known α, β .
- (b) Under the squared error loss, find the Generalized Bayes estimator of $p(1-p)$ with respect to the improper prior $[p(1-p)]^{-1}$, $0 < p < 1$.
- (c) Under the weighted squared error loss, with weight $w(p) = [p(1-p)]^{-1}$, obtain the Bayes estimator of p when the prior is $U(0, 1)$.
- (8 + 8 + 8 = 24 marks)**

7. Let X be a random variable with pdf $f(x)$, $x \in \mathbb{R}$.
- (a) Determine the MP- α test for testing

$$H_0 : f(x) = f_0(x) = \begin{cases} \frac{3}{64}x^2, & 0 \leq x \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{against} \\ H_A : f(x) = f_A(x) = \begin{cases} \frac{3}{16}\sqrt{x}, & 0 \leq x \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

- (b) Find the power of the test when $\alpha = 0.01$.

(5 + 5 = 10 marks)

8. Let $X_1, \dots, X_n$ be a random sample from the density $f_X(x) = \frac{1}{\theta}e^{-\frac{(x-\delta)}{\theta}}$, $x > \delta$.
If $\theta(> 0)$ is known, find an LR test of (exact) size α for testing:
- (a) $H_0 : \delta \leq \delta_0$ against $H_A : \delta > \delta_0$, for a fixed δ_0 ;
- (b) $H_0 : \delta = \delta_0$ against $H_A : \delta \neq \delta_0$, for a fixed δ_0 .

(7 + 5 = 12 marks)